

CLAUDE.md

CLAUDE.md

Field	Value
Revision	3
Created	2026-05-15
Last modified	2026-05-20
Status	active
Status summary	"Project status" section updated to reflect that the Go scaffold (5 modules, passing unit tests, Quickstart compose) has landed. New commands documented (go build/test). The pre-implementation language replaced with implementation-r1 reality.
Issues	none
Issues summary	—
Fixed	spec-path references (r2), pre-implementation-language update (r3)
Fixed summary	aligned with HRD-009/HRD-009b/HRD-013/HRD-014 landing in the same commit.
Continuation	bump again when first-implementation cycle completes HRD-010..HRD-012/HRD-016 live integrations.

Table of contents

- [INHERITED FROM Helix Constitution \(parent-discovery\)](#)
- [Project status](#)
- [Mission \(from the spec\)](#)
- [Intended stack](#)
- [Multi-host mirror convention](#)
- [Inheritance gate \(run before any commit touching root docs\)](#)
- [Spec-change rule \(load-bearing — docs/specs/mvp/specification.V3.md §"Specification documents"\)](#)
- [Project conventions from the spec \(apply when scaffolding\)](#)
- [constitutable/ directory \(parent-project extension hook\)](#)
- [Documentation artefacts \(PDF/HTML siblings\)](#)
- [Notes for future scaffolding](#)

This file provides guidance to Claude Code (claude.ai/code) when working with code in this repository.

INHERITED FROM Helix Constitution (parent-discovery)

Herald is consumed as a submodule of a parent project that already carries the Helix Constitution submodule at <parent>/constitution/. Herald therefore does **NOT** keep its own copy. Locate the constitution from any nested depth by walking up parents until you find <ancestor>/constitution/Constitution.md, or by running the canonical helper:

```
CONST_DIR="$(bash "$(find . -type d -name constitution -print  
-quit 2>/dev/null)/find_constitution.sh)"  
# or, more robustly, from any starting directory:  
CONST_DIR="$(bash <ancestor>/constitution/find_constitution.sh)"
```

For standalone development of Herald (no parent project), clone the constitution alongside Herald:

```
git clone git@github.com:HelixDevelopment/HelixConstitution.git \  
$(dirname "$PWD")/constitution
```

Once located, all rules in <discovered>/CLAUDE.md and the <discovered>/Constitution.md it references apply unconditionally. Herald-specific rules below extend them — they **MUST NOT** weaken any inherited rule. When this file disagrees with the discovered constitution, the constitution wins.

Canonical: <https://github.com/HelixDevelopment/HelixConstitution>

Read order on a cold start:

1. <discovered-constitution>/CLAUDE.md + Constitution.md — universal Helix rules. Inherited unconditionally.
2. <discovered-constitution>/AGENTS.md — agent guardrails (anti-bluff, no-guessing, paired mutations).
3. README.md — Herald overview + how-to.
4. This file (Herald-specific notes below).
5. docs/guides/HERALD_CONSTITUTION.md — Herald's project-specific constitutional extensions.
6. docs/guides/CONSTITUTION_INHERITANCE.md — operator/agent guide for the discovery contract + the inheritance gate.
7. docs/specs/mvp/specification.V3.md — mission spec (mostly TBD).

Project status

Herald is **pre-implementation**. As of this writing the repo contains:

- README.md — mission, deployment model, inheritance contract, quickstart.
- docs/specs/mvp/specification.V3.md — MVP spec stub (substantive sections TBD).

- docs/guides/HERALD_CONSTITUTION.md — Herald's project constitution (extends Helix).
- docs/guides/CONSTITUTION_INHERITANCE.md — operator/agent guide for parent-discovery + gate semantics.
- upstreams/ — Herald's mirror declarations (see below).
- tests/test_constitution_inheritance.sh + _meta.sh — paired inheritance gate (§1.1).
- .gitignore tuned for Go + .DS_Store.

Herald does **not** ship a constitution/ submodule of its own; the parent project provides it. See docs/guides/CONSTITUTION_INHERITANCE.md.

As of 2026-05-20 the Go scaffold has landed (first-implementation cycle r1). The repo now contains:

- go.work (gitignored locally; check in if the project wants reproducible workspaces — current convention: gitignored per spec §9.1).
- commons/ (L0) — commons/types.go with the full §11.0 Channel contract + Subscriber/CloudEventEnvelope/TraceContext/Branding/ChannelID; commons/clock.go Clock abstraction; commons/uuidv7.go; commons/branding.go per-flavor factory.
- commons_prefix/ — §8.2 3-letter prefix algorithm.
- commons_messaging/channels/null/ — full §11.14 null:// sandbox adapter (working, tested).
- commons_messaging/channels/tgram/ — Telegram adapter SCAFFOLD (HRD-011 open).
- commons_messaging/dispatch/claude_code/ — Claude Code session-resolution + envelope formatter; live claude --resume invocation pending (HRD-012 open).
- commons_storage/ — 9 SQL migrations (000001..000005) embedded via // go:embed; pgx + River + Redis wiring pending (HRD-010 open).
- pherald/cmd/pherald/ — Cobra CLI; pherald version works end-to-end; other subcommands stubbed with HRD-NNN error pointers.
- quickstart/ — Herald-specific Docker/Podman Compose + Dockerfile + otel-config + .env.example per §26.5 (HRD-008 open for operator validation). Migrated from containers/quickstart/ 2026-05-20 when the containers/ submodule was added.
- containers/ — git submodule pointing at git@github.com:vasic-digital/containers.git (the digital.vasic.containers Go module — runtime auto-detection + on-demand container boot + lifecycle/health). Imported by Foundation tests + the pherald doctor subcommand to bring Postgres + Redis + OTel up on-demand.

Build + test: from the repo root:

```
go build ./commons/... ./commons_prefix/... ./
    commons_messaging/... ./commons_storage/... ./
    commons_constitution/... ./commons_infra/... ./pherald/...
go test -race -count=1 ./commons/... ./commons_prefix/... ./
    commons_messaging/... ./commons_storage/... ./
    commons_constitution/... ./commons_infra/... ./pherald/...
```

Tests pass on Go 1.25+ (verified on 1.26). Workspace is configured via `go.work` listing 7 Herald modules (`commons`, `commons_constitution`, `commons_infra`, `commons_messaging`, `commons_prefix`, `commons_storage`, `pherald`). The 9 Helix-stack submodules under `submodules/` are referenced via `replace` directives in the consuming modules' `go.mod`, not via `go.work`.

Anti-bluff verification (run before any release tag or risky commit):

```
scripts/audit_antibluff.sh      # 3 invariants: §11.4 anchor +
                                tests + paired meta
scripts/codegraph_validate.sh   # CodeGraph index integrity (7
                                probes)
scripts/e2e_bluff_hunt.sh       # 14 end-to-end checks against
                                real services
```

`scripts/e2e_bluff_hunt.sh` is the canonical end-to-end smoke per Universal §11.4. It builds `pherald`, runs the full test suite, starts a real Gin server + hits every `/v1` route + asserts response bodies, boots a real Postgres container via the `containers/` submodule, runs the M2 integration tests against it, and graceful-shutdowns. ALL 14 invariants must PASS — a single FAIL means a real feature is broken for end users. At 2026-05-20 it reports 14/14 PASS.

When the user asks to "add a feature" the spec is the source of truth — find the relevant §, then the relevant module + package, then the relevant HRD-NNN if one is already open. New work opens a new HRD-NNN in `docs/Issues.md` per V3 §8.3 lifecycle.

Do not invent build/test commands beyond what `go test ./<module>/...` provides. Live-integration tests (Telegram bot, Claude Code session, real Postgres) require operator-supplied credentials — see `docs/CONTINUATION.md` for the live-test handoff prompt.

Mission (from the spec)

Ingesting system events and reliably fanning them out to multiple notification channels so every alert reaches the right destination without confusion.

The spec mentions "Integration into the Constitution" as a planned section — Herald is intended to plug into a larger governance/policy system that lives outside this repo. Treat that as a real constraint when designing interfaces, even though the details are TBD.

Intended stack

Go, inferred from `.gitignore` (`*.test`, `go.work`, `go.work.sum`, `coverage.*`, `profile.cov`). When you scaffold, default to standard `cmd/` + `internal/` layout unless the user asks otherwise, and use `go test ./... /go build ./... /go vet ./...` — there is no project-specific tooling to override these yet.

Multi-host mirror convention

upstreams/ contains one script per mirror host:

- GitHub.sh — primary: git@github.com:vasic-digital/Herald.git
- GitLab.sh — git@gitlab.com:vasic-digital/herald.git
- GitFlic.sh — git@gitflic.ru:vasic-digital/herald.git
- GitVerse.sh — git@gitverse.ru:vasic-digital/Herald.git

Each script exports a single UPSTREAMABLE_REPOSITORY variable and is meant to be **sourced** (`. upstreams/GitHub.sh`), not executed for its output. The naming is intentional — capitalization matches the host's brand (GitFlic, GitVerse). Don't normalize these to lowercase or collapse them into one file; the per-host split is the interface other tooling (likely external mirror-push scripts) keys on.

When adding a new mirror, copy an existing script and change only the URL — keep the `#!/bin/bash` shebang, blank line, and export form identical so any consumer that greps these files keeps working.

Inheritance gate (run before any commit touching root docs)

```
bash tests/test_constitution_inheritance.sh          # 6 invariants
              (I1-I6), 9 checks
bash tests/test_constitution_inheritance_meta.sh     # paired §1.1
              mutation proof
```

The gate inline-walks parents for <ancestor>/constitution/Constitution.md. I5 is split into I5a–d (one check per root doc that must declare parent-discovery: CLAUDE.md, AGENTS.md, docs/guides/HERALD_CONSTITUTION.md, README.md). I6 forbids re-introducing <repo-root>/constitution/ or .gitmodules — the §104 invariant.

If either script fails, fix at root cause per Universal §11.4.4. Never silently accept the FAIL.

Spec-change rule (load-bearing — docs/specs/mvp/specification.V3.md §"Specification documents")

Any modification to a file under docs/specs/ (any depth) triggers **mandatory comprehensive planning and implementation** of the implied changes — you may not edit the spec in isolation. This rule does **not** apply to creating or renaming files; for those, ask the operator what to do with the new path. Treat every spec edit as a project-wide ripple, not a doc tweak.

Project conventions from the spec (apply when scaffolding)

These are declared in docs/specs/mvp/specification.V3.md and are easy to miss because no code enforces them yet:

- **Workable-item prefix:** HRD- (e.g. HRD-001). Use it for issues, status entries, fix logs.
- **Flavor binaries:** each Herald flavor ships as its own CLI binary, named `<prefix>herald` — pherald (Project Herald), sherald (System Herald), etc. Designed for CI / pipeline / cron / AI-agent invocation.
- **Layered shared code:** commons → commons_messaging (level 1) → ... → flavor. Put new shared code in the **lowest layer that still makes sense**; flavors inherit upward. commons_messaging owns the Telegram / Max / Slack / Email / Markdown-export integrations.
- **Messenger integration priority:** Telegram → Max → Slack (then Email, then Markdown/PDF/HTML export). Microsoft Teams, Lark, Discord, WhatsApp, Viber are explicitly later iterations — don't pre-implement.
- **Conversation diary:** every message in/out is appended to docs/herald/diary/main.md and re-exported to main.pdf + main.html in sync. Don't break this invariant when designing channel I/O.
- **Container stack:** Postgres (main DB) + Redis (in-memory) bundled via the containers submodule (<https://github.com/vasic-digital/containers>). All container names start with herald; all host ports start with 70XXX (70001, 70002, ...) to avoid collisions.
- **Credentials:** .env (git-ignored) with a committed .env.example. Resolution order: exported shell vars from .bashrc/.zshrc load first, then .env overrides them on key collision.
- **Vendored SDKs:** any official/unofficial messenger SDK or API client we depend on goes in as a **git submodule**, e.g. commons_messaging/sdk/telegram or commons_messaging/api/telegram — not go get'd into go.mod.

constitutable/ directory (parent-project extension hook)

The empty constitutable/ directory at the repo root is intentional. Per the spec, a parent project may drop additional Constitution.md / CLAUDE.md / AGENTS.md (in constitutable/, constitutable/<flavor>/, constitutable/<flavor>/<variant>/, etc.) to layer extensions or overrides on top of the discovered constitution/ submodule. Apply-order: constitution/ submodule → constitutable/ extensions → Herald's own docs. Do not delete the directory because it's empty.

Documentation artefacts (PDF/HTML siblings)

docs/guides/HERALD_CONSTITUTION.md and docs/guides/CONSTITUTION_INHERITANCE.md each ship with a committed .pdf sibling. When you edit one of these Markdown files, the PDF goes stale — flag it; do not regenerate silently unless the operator asks.

Notes for future scaffolding

- The repo is in main branch and committed under "Milos Vasic" — no other contributors yet.
- `.claude/` exists but is empty; project-local Claude config can go there.
- `LICENSE` is present (do not overwrite without asking).
- `.DS_Store` is now git-ignored; do not re-add the previously-stray files.